

TANKHOY, Russian Federation

WMO: 308240

Lat: 51.555N

Lon: 105.113E

Elev: 1549

StdP: 13.89

Time Zone: 8.00 (E08)

Period: 94-06

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-20.9	-15.1	-23.9	1.5	-20.6	-18.5	2.1	-14.5	14.2	10.6	11.4	14.9	3.6	180	0.656

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.1	72.6	63.1	69.8	61.9	67.5	60.9	65.3	69.9	63.4	68.0	61.8	66.2	4.8	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.5	92.7	68.5	61.5	86.3	66.2	59.8	81.4	64.7	31.2	69.7	29.7	68.4	28.5	66.4	72.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.4	13.7	11.5	-24.4	79.5	8.1	4.2	-30.2	82.5	-34.9	85.0	-39.4	87.3	-45.3	90.4
			-24.1	68.1	8.2	2.6	-30.0	70.0	-34.8	71.5	-39.4	73.0	-45.4	74.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	33.1	4.8	2.9	15.7	33.0	44.1	53.0	60.4	58.3	47.9	36.2	24.3	15.1	
	DBStd	20.69	9.25	10.34	10.41	6.64	6.84	5.51	4.67	3.94	5.21	6.66	7.96	6.16	
	HDD50	6916	1403	1319	1062	511	211	30	0	1	100	428	770	1081	
	HDD65	11646	1868	1739	1527	961	649	362	157	212	513	892	1220	1546	
	CDD50	762	0	0	0	0	27	119	321	257	37	1	0	0	
	CDD65	17	0	0	0	0	0	1	13	3	0	0	0	0	
	CDH74	54	0	0	0	0	4	7	41	2	0	0	0	0	
	CDH80	5	0	0	0	0	0	0	5	0	0	0	0	0	
Wind	WSAvg	5.0	4.8	3.7	4.4	5.0	5.2	4.7	4.2	4.6	5.4	5.7	5.9	5.9	
Precipitation	PrecAvg	30.7	0.9	0.4	0.7	1.9	2.1	3.9	5.8	6.3	4.0	1.6	1.6	1.5	
	PrecMax	38.5	2.0	0.7	1.5	3.9	3.6	9.4	10.7	9.7	8.5	3.0	3.6	2.3	
	PrecMin	24.6	0.4	0.1	0.3	0.6	0.9	0.6	2.2	3.9	1.1	0.3	0.8	0.4	
	PrecStd	4.5	0.4	0.2	0.3	1.0	0.8	2.0	2.5	1.9	2.0	0.6	0.7	0.6	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.0	32.2	45.8	57.2	71.7	74.2	78.6	72.5	62.9	59.1	44.8	37.2	
		MCWB	28.9	26.6	36.2	43.9	54.2	58.8	66.0	63.5	55.1	46.9	37.8	31.5	
	2%	DB	27.5	27.1	39.0	50.4	65.0	69.1	74.2	69.9	60.8	51.9	41.0	31.6	
		MCWB	25.0	23.4	32.3	40.6	51.4	58.2	65.5	62.7	55.6	45.1	36.0	28.4	
	5%	DB	22.0	22.9	35.2	45.6	60.2	65.1	71.3	68.1	59.1	48.4	38.2	28.0	
		MCWB	19.9	20.5	30.4	37.9	48.7	56.7	63.7	62.3	54.2	43.4	34.1	25.4	
	10%	DB	18.2	19.2	32.0	41.9	56.3	62.2	68.6	66.0	57.1	45.9	35.3	24.8	
		MCWB	16.5	17.3	28.2	35.6	46.8	55.1	62.2	60.9	52.4	41.7	32.1	22.6	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	29.6	28.0	36.5	45.6	56.4	62.0	68.9	66.3	59.1	48.0	38.8	32.8
			MCDB	31.4	31.8	44.5	53.7	69.2	68.9	74.7	69.6	60.8	54.9	43.0	34.7
2%		WB	25.4	23.5	33.3	41.1	53.1	60.0	66.8	64.3	56.6	46.1	36.5	28.5	
		MCDB	27.7	26.7	38.1	50.1	61.7	66.0	72.4	68.3	59.8	50.8	40.3	31.6	
5%		WB	20.1	20.5	30.6	38.7	50.5	58.3	64.6	62.8	54.7	43.8	34.5	25.4	
		MCDB	21.9	22.8	34.6	44.7	58.3	63.5	69.6	66.8	58.3	48.0	37.6	27.7	
10%		WB	16.5	17.3	28.5	36.3	47.7	56.5	62.9	61.4	52.9	41.7	32.4	22.8	
		MCDB	18.1	19.2	32.0	41.2	55.0	61.2	67.7	65.2	56.4	45.5	35.4	24.6	
Mean Daily Temperature Range	5% DB	MDBR	15.6	19.0	19.3	14.0	16.0	14.0	12.1	11.3	12.4	12.6	13.0	13.8	
		MCDBR	18.0	22.1	22.1	20.6	23.0	19.5	17.1	14.3	14.9	16.5	15.9	16.9	
		MCWBR	15.1	18.5	16.0	12.7	12.5	10.9	9.4	8.8	10.1	11.6	12.5	13.7	
	5% WB	MCDBR	16.8	21.8	20.2	19.2	21.1	15.9	14.9	12.6	13.5	15.8	15.3	16.0	
		MCWBR	15.3	18.5	15.2	12.2	12.8	10.1	9.3	8.8	10.2	12.1	12.3	13.4	
Clear-Sky Solar Irradiance	taub		0.274	0.301	0.342	0.391	0.456	0.451	0.440	0.402	0.346	0.322	0.292	0.275	
	taud		2.100	2.075	2.099	2.167	2.036	2.176	2.259	2.347	2.454	2.400	2.179	2.070	
	Ebn at Noon		226	254	266	267	253	256	257	262	267	250	218	201	
	Edh at Noon		24	33	40	42	51	45	41	35	28	24	22	21	
All-Sky Solar Radiation	RadAvg		408	739	1152	1458	1763	1839	1704	1434	1127	705	402	287	
	RadStd		19	39	62	101	95	109	117	92	112	54	27	21	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page